

Entity Extraction Pipeline

Objective

Convert industrial text into structured entities.

Workflow

```
graph TD; A[OCR Text] --> B[Chunking]; B --> C[NER Model]; C --> D[LLM Extraction]; D --> E[Entity Validation]; E --> F[Relationship Extraction]; F --> G[Knowledge Graph Builder];
```

Entity Types

Equipment

Pump P101

Motor M201

Valve V301

Failure

Bearing Failure

Leakage

Overheating

Incidents

Shutdown Event

Near Miss

Fire Event

Regulations

OISD

ISO 9001

Factory Act

Personnel

Engineer

Operator

Technician

Example

Input

Pump P101 experienced bearing failure on 10 Jan 2025.

Output

```
{
  "equipment": "Pump P101",
  "failure": "Bearing Failure",
  "date": "2025-01-10"
}
```

Relationship Extraction

Pump P101

↓

FAILED_IN

↓

Bearing Failure

↓

CAUSES

↓

Overheating

Models

spaCy

Gemini

LangExtract

GLiNER